

PRAVEEN KUMAR S

AI & Backend Systems | Agentic Systems | RAG | Forward Deployed Engineering
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ACHIEVEMENTS

2× Winner: SAMARTHA’25 National Hackathon (INR 1.5L); NALSAR University AI Trading Challenge (solo, INR 30K)
10× National Finalist: Top 10 at Volkswagen i-Mobilothon 5.0, IntelliSmart INSTINCT, and ISMA SugarNXT
50+ hackathons: Participated across national competitions, leading problem framing, technical direction, rapid prototyping, and final delivery

PROJECTS

1. ARIV – Agentic Payment-Recovery Control Plane | [GitHub](#) | Sep. 2026

Python, FastAPI, SQLAlchemy, PostgreSQL, Redis, Qdrant, Next.js, Docker, Pytest, Testcontainers

Benchmark / verified scope: 43 Razorpay test-mode cases (INR 1.81L at risk); 302 tests; 29/30 actions completed; 9/18 non-retryable failures halted; 10 seeds × 10,000 synthetic cases.

- Built ARIV so **AI proposes, PolicyEngine authorizes, and revenue counts only after Razorpay confirmation**; decisioned all 43 Razorpay test-mode cases (INR 1.81L at risk) and attributed one confirmed INR 100 test-mode payment.
- Ran and published a 10-seed × 10,000-case ablation; economic ranking and route intelligence drove modeled gains while LLM/Qdrant added none, validating an advisory-only AI architecture.
- Hardened execution with idempotent actions, optimistic locking, and event-order validation to prevent duplicate actions on replayed or out-of-order webhooks; passed **302 tests**.
- Classified provider failures and halted **9 of 18 non-retryable failure scenarios** instead of blindly retrying; completed **29/30 test-mode provider actions**.

2. Info-i – Multimodal Fact-Verification System | [GitHub](#) | Jan. 2026

Python, FastAPI, LangGraph, Qdrant, Redis, OCR/ASR, React, Docker

Benchmark / verified scope: Three verdict classes; three non-text input types; cited evidence, confidence scoring, and human-review routing.

- Built a fact-verification pipeline with LangGraph, live SERP evidence, and Qdrant retrieval, returning three verdict classes—True, False, or Mixture—with confidence and cited sources.
- Extended verification to three non-text inputs—audio, screenshots, and images—using OCR/ASR and EXIF/noise forensics, with uncertain cases routed to human review.

3. SodhaneGPT – Explainable Crime-Intelligence Data Platform | [GitHub](#) | Jul. 2026

Python, FastAPI, Pandas, Parquet, XGBoost, scikit-learn, NetworkX

Benchmark / verified scope: 100K+ synthetic-record design target; three analysis layers; reproducible train/validation/test splits.

- Built a reproducible ML data platform with schema validation, Parquet versioning, and fixed train/validation/test splits, designed for a 100K+ synthetic-record target.
- Combined three analysis layers—anomaly detection, spatial clustering, and graph centrality—to return a why/evidence/confidence trace with each composite risk score.

4. Hazard_Det – Privacy-First Edge Road Intelligence | [GitHub](#) | Nov. 2025

Python, YOLOv8, OpenCV, Kalman tracking, RabbitMQ, Raspberry Pi

Benchmark / verified scope: Plate model trained for 50 epochs at 640 px; three event-context sources: GPS, IMU, and CAN.

- Built an edge-to-cloud hazard pipeline with YOLOv8, Kalman tracking, and RabbitMQ, de-duplicating detections and publishing severity-scored geospatial events.
- Protected footage before upload with face/plate anonymization and a plate detector trained for 50 epochs at 640 px; attached GPS, IMU, and CAN context to each event.

EXPERIENCE

AI & Application Development Intern | Jun. 2025 – Aug. 2025

BillionBright Solutions LLP

Remote

- Built image-generation workflows across three model paths—Stable Diffusion, LoRA, and Fooocus—connecting orchestration and inference to application interfaces.
- Standardized enhancement, orchestration, and deployment steps across the three paths, producing a repeatable release workflow.

TECHNICAL SKILLS

Primary: Python, SQL, FastAPI, Pydantic, SQLAlchemy (async), PostgreSQL, Redis, Pytest, Testcontainers

Agentic AI & Retrieval: LangGraph, RAG, Qdrant, embeddings, LLM tool orchestration

Applied ML: scikit-learn, XGBoost, YOLOv8, OpenCV, Pandas

Product & Delivery: TypeScript, Next.js, React, Docker, Docker Compose, RabbitMQ, Parquet, Git

EDUCATION

Sri Sairam Institute of Technology

| Chennai, TN

B.Tech, Artificial Intelligence & Data Science | CGPA: 7.17/10

| Sep. 2023 – May 2027 (Expected)